

WHAT LIQUIFIES A HYDROGEL?

Variable Investigation - Temperature, Acidic Conditions, and Basic Conditions

THE CHALLENGE

Gel-like products such as hand sanitizer can spread and become much thinner when they leave a bottle and reach your hand. In this activity, your team will test ONE possible factor that may change the physical state of a model hydrogel. Your results will be combined with the rest of the class to determine which conditions have the strongest effect.



SCAN FOR THE MANUAL

How the class is organized

Teams	Variable	Student 1	Student 2	Student 3
Teams 1-4	Temperature	10 C	Room temperature	37 C
Teams 5-7	Acidic pH	pH 5	pH 4	pH 3
Teams 8-10	Basic pH	pH 9	pH 10	pH 11

Safety

- Wear splash goggles throughout the experiment. Wear gloves when directed by your instructor.
- Do not touch or taste acidic or basic test solutions or treated hydrogel.
- Do not handle dry LiquiLock powder. Your instructor has already pre-portioned it in your sample cup.
- Temperature teams: heat only water in a beaker. Never place a plastic cup on a hot plate.
- If a solution spills, a cup cracks, or you are unsure of a step, stop and ask.
- Bulk hydrogel goes in the lined trash container - not down the sink.

Per-team materials

Team type	Materials
All teams	3 clear plastic sample cups, each preloaded by the instructor with 0.50 g LiquiLock hydrogel polymer; 3 spoons; graduated cylinder(s); room-temperature DI water; timer; goggles
Temperature	Thermometer; access to shared cold-water bath; beaker + hot plate for warm-water condition
Acidic pH	Prepared pH 5, pH 4, and pH 3 solutions in labeled small cups, separate from sample cups; pH paper/indicator as directed
Basic pH	Prepared pH 9, pH 10, and pH 11 solutions in labeled small cups, separate from sample cups; pH paper/indicator as directed

Polymer used: sodium polyacrylate water-absorbing polymer (LiquiLock). Each student uses one preloaded cup and tests one assigned condition.

STUDENT PROCEDURE

Each student runs one condition. Record your team's results on the last page.

FAIR TEST RULE

Only your assigned temperature or pH should change. Hydrogel amount, water volume, stirring, and waiting time must stay the same for all three students.

Common sample procedure - everyone does this

- 1 Take one clear plastic cup preloaded with 0.50 g LiquiLock hydrogel polymer. Do not handle the dry powder.
- 2 Label your sample cup with your name and assigned condition.
- 3 Add 40 mL room-temperature DI water to your sample cup; stir gently with a spoon.
- 4 Allow the hydrogel to hydrate for 5 minutes.
- 5 Obtain your assigned test liquid. Measure and record its temperature or pH before adding it to the gel.
- 6 Measure 25 mL of your assigned test liquid and add it to your hydrated sample.
- 7 Stir each sample exactly 10 strokes using the same motion.
- 8 Wait 5 minutes.
- 9 Score the hydrogel from 1 to 5, note whether free water is visible, and write one short observation.

Variable-specific instructions

TEMPERATURE TEAMS

Student 1: 25 mL DI water at 10 C.
Student 2: 25 mL DI water at room temperature.
Student 3: 25 mL DI water at 37 C.
 Use a thermometer; record the water temperature before addition.

ACID TEAMS

Student 1: 25 mL pH 5 solution.
Student 2: 25 mL pH 4 solution.
Student 3: 25 mL pH 3 solution.
 Use the labeled small solution cups.
 Confirm/record pH as directed before addition, not in the gel.

BASE TEAMS

Student 1: 25 mL pH 9 solution.
Student 2: 25 mL pH 10 solution.
Student 3: 25 mL pH 11 solution.
 Use the labeled small solution cups.
 Confirm/record pH as directed before addition, not in the gel.

Hydrogel rating scale

Score	Description	What it looks like
5	Fully solid gel	No obvious change
4	Slightly smaller / softened	Small change
3	Partially liquified	About half gel-like / half liquid
2	Mostly liquid	Only a small amount of gel structure remains
1	Fully liquid	Gel structure is no longer apparent

OBSERVATION TIP

Also record whether you can see free water released from the gel. Two samples can receive the same 1-5 score but still look different.

HYDROGEL VARIABLE TEST

Data Sheet

ONE TEAM / ONE VARIABLE / THREE CONDITIONS

STUDENT NAMES _____

ASSIGNED VARIABLE

☐

Temperature

☐

Acidic pH

☐

Basic pH

TEMPERATURE

10 C cold | room temp | 37 C warm

ACIDIC pH

pH 5 | pH 4 | pH 3

BASIC pH

pH 9 | pH 10 | pH 11

Same procedure for every sample

- 1 Your cup is preloaded with 0.50 g LiquiLock. Add 40 mL room-temperature DI water.
- 2 Stir gently; hydrate for 5 minutes. Do not handle dry powder.
- 3 Measure and record the test liquid's temperature or pH, then add 25 mL to the gel.
- 4 Stir exactly 10 strokes. Wait 5 minutes.
- 5 Score the gel, mark whether free water is visible, and write observations.

Hydrogel score

5 solid / no change**4** softened**3** partly liquefied**2** mostly liquid**1** fully liquid

Record your three samples

Sample	Target	Test liquid temp / pH	Score 1-5	Free water?	Observations
1			_____	YES / NO	
2			_____	YES / NO	
3			_____	YES / NO	

Quick conclusion

1. Which of your three conditions changed the gel the most?

2. What evidence supports your answer?

3. Would your variable alone explain why a sanitizer-like gel changes when it reaches a hand? Why or why not?

CLEANUP

Put used hydrogel in the lined trash container - do not put bulk gel down the sink.

Rinse reusable glassware/spoons only after the bulk gel has been removed.